

GENERAL SPECIFICATION G7.13

Two-Way Radio Communications
(Walkie-Talkie) System

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G7.13 TWO-WAY RADIO COMMUNICATIONS (WALKIE-TALKIE) SYSTEM

G7.13.1 General

(a) References

(i) The following is a list of standards which may be referenced in this Section:

- 1) Electronic Industry Association (EIA) and Telecommunications Industry Association (TIA)

Standard	Subject
EIA/TIA 568-A	Commercial Building Telecommunications Cabling Standard
EIA/TIA 569-A	Commercial Building Standard for Telecommunications Pathways and Spaces
EIA/TIA 607	Commercial Building Grounding and Bonding Requirements for Telecommunications
EIA/TIA TSB 67	Transmission Performance Specification for Field Termination of Unshielded Twisted Pair Cabling Systems

- 2) International Electrotechnical Commission (IEC / EN)

Standard	Subject
60529	Degrees of Protection Provided by Enclosures (IP Code)
EN 300 086 and 300 113	Electromagnetic compatibility and Radio spectrum Matters (ERM)

- 3) European Telecommunications Standard Institute (ETSI)

Standard	Subject
ETS 300 086 and 300 113	Electromagnetic compatibility and Radio spectrum Matters (ERM)
TR 102 398	Digital Mobile Radio (DMR) General System Design
TS 102 361	Digital Mobile Radio (DMR) Systems
TS 102 362	Conformance testing for the Digital Mobile Radio (DMR)

Standard	Subject
TS 102 490	Peer-to-Peer Digital Private Mobile Radio using FDMA with a channel spacing of 6,25 kHz with e.r.p. of up to 500 mW
TR 102 335-1	System reference document for harmonized use of Digital Mobile Radio (DMR); Part 1: Tier 1 DMR#, expected to be for general authorization with no individual rights operation
TR 102 335-2	System reference document for harmonized use of Digital Mobile Radio (DMR); Part 2: Systems operating under individual licences in the existing

- 4) National Electrical Manufacturers Association (NEMA): 250, Enclosures for Electrical Equipment (1,000 Volts Maximum).
- 5) National Fire Protection Association (NFPA): 70, National Electrical Code (NEC).
- 6) SPRING Singapore

Standard	Subject
CP 5	Electrical Installation
SS 551	Code of Practice for earthing
SS 555	Code of Practice against lightning protection
IDA COPIF	Code of Practice for Info-Communications Facilities in Building
IDA TS EMC	EMC Requirements for Telecommunication Equipment
IDA TS L1-L3	Booklet of Specifications for Telecommunication Cables and Ancillary Accessories
IDA CP L1	Code of Practice for Internal Telecommunication Wiring

7) Underwriters Laboratories, Inc. (UL)

Standard	Subject
1459	Telephones
1863,	Wires and Jacks

(b) Compliance

- (i) The complete telecommunications installation shall be compliant with EIA/TIA 568-A and IDA COPIF; and
- (ii) Applicable ETSI, IEC and EN standards.

G7.13.2 Products

(A) General

(i) General

- 1) The Emergency Two Way Radio Communications System shall be Digital, Time Division Multiple Access (TDMA), Voice, GPS and with data application.
- 2) The system shall have the following features:
 - Voice and data GPS (without Internet connection).
- 3) The Two Way Radio Communications System shall comprise of necessary Base Stations, Repeater Stations, Wireless Bridges, low loss coaxial cables, Power Splitters, Vocoder, Antenna, etc. as required for safe, reliable and fully functional system.

(ii) Base Stations

- 1) The contractor shall provide necessary base stations to ensure satisfactory communications system. The base stations shall be compatible to the existing system;
- 2) It shall be able to work in conjunction with software to perform GPS tracking for handset;
- 3) It shall be Digital, TDMA (Time Division Multiple Access) technology, able to support 2 time slots for both voice & data, GPS, Voice Logging & Lone Worker features;
- 4) Complete with individual single charger unit;
- 5) It shall have Display, GPS model at UHF/VHF;
- 6) The Transmitter shall have 4FSK (4 level Frequency Shift Key) Digital FM Modulation, Digital Vocoder type AMBE+2 (Advance Multi Band Excitation), Digital Protocol (ETSI = TS102 361-1);
- 7) The Battery Life shall be 9hrs /13 hrs (analog/digital);
- 8) GPS horizontal accuracy to within 10 meters; and
- 9) Transmitter Power shall be 1W/5W (Low/High) at 7.5 volt nominal.

(iii) Repeater Stations

- 1) Supports 2 simultaneous voice or data paths in digital TDMA mode;
- 2) Connect allows networks to existing base stations and / or repeaters to expand voice and data coverage;
- 3) Rack or wall mountable or desktop housing as approved by S.O.;
- 4) Doubles the number of users you can have on a single licensed 12.5 kHz channel;
- 5) Monitor and manage repeater via the optional diagnostic and control utility;
- 6) Automated battery back-up, battery to be included;
- 7) Integrates data and voice to increase operational efficiency;
- 8) Integrated power supply and 100% continuous full duty cycle at high power;
- 9) Scalable, single-site digital solution that enables a large group of radio users to share both voice and data on the same system;
- 10) Operates in analog or digital mode;
- 11) Conforms to EC199/5/EC (R&TTE), EN 300 086, En 300 113;
- 12) UHF/VHF – 16 Channels, 403-490Mhz, 450-512 Mhz, 136-174Mhz;
- 13) 230vac 50 Hz, -30 to +60 degrees Centigrade;
- 14) Transmitter – FM modulation, 4FSK Digital Modulation, Digital Vocoder type; and
- 15) (AMBE+2), Digital Protocol (ETSI=TS102 361-1).

(iv) Wireless Bridge

- 1) The contractor shall provide the necessary numbers of Wireless Bridge so as to integrate with the existing system; and
- 2) Contractor shall provide necessary power and lightning protection.

(v) Portable Handset Radio

- 1) IP57 submersibility and Intrinsically Safe (used in locations where flammable gases, vapors or combustible dust may be present;
- 2) Digital, TDMA technology, able to support 2 time slot for both voice & data, GPS, Voice Logging & Lone Worker features;
- 3) Supports integrated applications including Text Messaging Services and Location Services;
- 4) Enables additional functionality including dispatch data, and enhanced call signalling;
- 5) Enhanced call management features that shall include call alert, emergency, remote monitor, push-to-talk ID, radio check, private call, all call, and radio disable;
- 6) Complete with individual single charger unit;

- 7) Display, GPS model at UHF/VHF;
 - 8) Transmitter – FM modulation, 4FSK Digital Modulation;
 - 9) Digital Vocoder type (AMBE+2);
 - 10) Digital Protocol (ETSI=TS102 361-1);
 - 11) Battery / Battery Life: 2200 mAh Li-Ion, 9 hrs /13 hrs (analog/digital);
 - 12) GPS horizontal accuracy to within 10 meters; and
 - 13) Transmitter Power: 1W/5W (Low/High), 7.5 volt nominal.
- (vi) Cables
- 1) All power, control, data, low loss coaxial cables, etc. shall be FR (Fire Resistant) type. All cables shall be installed in cable trays in accordance with specifications.
- (vii) Antenna
- 1) All necessary antenna and associated accessories for connection shall be provided for the entire communication system.
- (viii) Multi-Unit Charger
- 1) Multi-Unit Charger (for simultaneous charging of 6 nos. of handsets), UK Plug, one unit, 230 V AC, 50 Hz supply shall be suitable for long term battery charging.
- (ix) Power Splitter
- 1) The contractor shall provide number of power splitters along with amplifiers and antenna as necessary for fully functional communication system.
- (x) Software and Licenses
- 1) The software for all equipments including the OS shall be provided along with their licenses for design, configuration, programming, and troubleshooting for GPS, Lone Worker, Remote Monitoring, Voice logging and Text messaging.
- (xi) UPS and Battery backup
- 1) All the communication equipments shall be powered from UPS with a battery backup of two hours. The UPS and battery shall be mounted in a proper enclosure in accordance with the specifications.
- (xii) Factory Mutual Approvals
- 1) Portable radios shall be certified by FM Approvals in accordance with Canada and U.S. Codes as intrinsically safe for use in Class I, II, III, Division 1, Groups C,D,E,F,G, or equivalent IEC zones when properly equipped with a FM approved battery option. They shall also be approved for use in Class I, Division 2, Groups A, B, C, D. or equivalent IEC zone classification.
- (xiii) Environmental Specifications
- 1) Operating Temperature : -30° C / +60° C
 - 2) Storage Temperature : -40° C / +85° C

- | | | |
|----|-----------------|--|
| 3) | Thermal Shock | : Per MIL-STD |
| 4) | Humidity | : Per MIL-STD |
| 5) | ESD | : IEC-801-2KV |
| 6) | Water Intrusion | : IEC 60529 - IP57 |
| 7) | Packaging Test | : MIL-STD 810D and E |
| 8) | Conforms to | : EC 1999/5/EC (R&TTE - Radio and
Telecommunications Terminal
Equipment), EN 300 086, EN 300 113 |

(B) Cables

(i) General

- 1) Cables shall meet flame propagation test requirements of IEC 60332, Category C; and
- 2) Cables shall be low smoke halogen free meeting all applicable requirements of IEC 61034, IEC 60754-1 and BS 6724.

(ii) Type 20, Telephone Station Cables

- 1) General: Cables shall be suitable for telephone system applications;
- 2) Conductors:
 - Solid copper, 0.5 mm²; and
 - Insulation: Colour coded polyethylene.
- 3) Cabling:
 - Unshielded twisted pairs;
 - Category 5;
 - Number of pairs: Four twisted pairs; and
 - Jacket: Low smoke halogen free.

(iii) Type 21, Data System Cables

- 1) General: Cables shall be suitable for data system applications;
- 2) ETL verified to TIA 568 TSB;
- 3) Conductors:
 - Solid copper, 0.5 mm²; and
 - Insulation: Colour coded polyethylene.
- 4) Cabling:
 - Unshielded twisted pairs;
 - Category 5;
 - Number of pairs: Four twisted pairs; and
 - Jacket: Low smoke halogen free.

(C) Connectors

(i) Patch Panels

- 1) General: Exceed EIA/TIA 568-A and IDA COPIF standards for Category 5 service;
- 2) High density RJ-45 receptacles, number as shown on the Drawings;
- 3) Wiring type: 568B;
- 4) Contacts:
 - Beryllium copper with 50-microinchgold plating; and
 - Type 110 contacts with tin-lead plated insulation displacement contacts (IDC).
- 5) Paired punch sequence that allows a pair twist within 12 mm of termination[
- 6) Integral cable management bar; and
- 7) Wallmount bracket or 19 inch rackmount as shown or required for application.

(ii) Block Terminals

- 1) General: Exceed EIA/TIA 568-A and IDA COPIF standards for Category 5 service;
- 2) Contacts:
 - Beryllium copper with 50-microinchgold plating; and
 - Type 110 contacts with tin-lead plated insulation displacement contacts (IDC).
- 3) Markings to designate Tip and Ring and colour-coded pairs.

(iii) Distribution Point

- 1) General: Exceed EIA/TIA 568-A and IDA COPIF standards for Category 5 service;
- 2) High density, number as shown on the Drawings;
- 3) Wiring Type: 568B;
- 4) Contacts:
 - Beryllium copper with 50-microinchgold plating; and
 - Type 110 contacts with tin-lead plated insulation displacement contacts (IDC).
- 5) Paired punch sequence that allows a pair twist within 12 mm of termination;
- 6) Integral cable management bar; and
- 7) Wallmount bracket or 19 inch rackmount as shown or required for application.

(iv) Enclosure: IP 55 (minimum).

- (v) Telephone Jacks
 - 1) General:
 - Exceed EIA/TIA 568-A and IDA COPIF standards for Category 5 service.
 - 2) Jack Type: RJ45;
 - 3) Wiring Type: 568B;
 - 4) Contacts:
 - Beryllium copper with 50-microinchgold plating; and
 - Type 110 contacts with tin-lead plated insulation displacement contacts (IDC).
 - 5) Paired punchdown sequence; and
 - 6) Colour: White.
- (vi) Data Jack
 - 1) General: Exceed EIA/TIA 568-A and IDA COPIF standards for Category 5 service;
 - 2) Jack Type: RJ45;
 - 3) Wiring Type: 568A;
 - 4) Contacts:
 - Beryllium copper with 50-microinchgold plating; and
 - Type 110 contacts with tin-lead plated insulation displacement contacts (IDC).
 - 5) Paired punchdown sequence; and
 - 6) Colour: Blue.
- (D) Outlet Jack Frame
 - (i) Designed to fit standard outlet box;
 - (ii) Two jacks per frame; and
 - (iii) Colour: Telco ivory.
- (E) Wall Plates
 - (i) Meet applicable requirements of Article Device Plates in Section 16050, Basic Electrical Materials and Methods.

G7.13.3 Execution

- (A) General
 - (i) Install all components in accordance with manufacturer's recommendations.
 - (ii) The complete telecommunications system installation shall be compliant with EIA/TIA 568-A and IDA COPIF.
 - (iii) Each wiring system shall be installed in a star topology.

- (iv) The vendor shall provide all incidental materials required for a complete installation.
- (v) All cables, terminations and jacks shall be labelled.
- (B) Cables
 - (i) Cable installation shall meet applicable requirements of EIA/TIA or SS/IEC or Article 800 of the NEC.
 - (ii) Install cables in conduits and other pathways that have been installed in accordance with the requirements of EIA/TIA 569-A.
 - (iii) Cables installed exposed in plenum spaces shall be plenum rated.
 - (iv) Terminate cables on 110 style contacts using approved punchdown tools.
 - (v) Pair twisting to be maintained to within 12 mm of termination.
 - (vi) All cables shall be secured at patch panels, distribution points and block terminals.
- (C) Connectors
 - (i) Install patch panels for all data cable systems.
 - (ii) Install block terminal and distribution points for all telephone systems.
 - (iii) Provide 100 percent spare ports and connection blocks.
 - (iv) Install patch panels and block terminals with the long axis vertical unless otherwise indicated.
 - (v) Install patch panels and block terminals in IP 55 enclosures unless otherwise indicated.
- (D) Earthing
 - (i) In accordance with EIA/TIA 607 and IDA COPIF; and
 - (ii) SS 551.
- (E) Testing
 - (i) General
 - 1) All testing is to conform to EIA/TIA TSB 67 specifications for testing copper wire.
 - 2) All data cables shall be tested to Category 5 performance standards.
 - 3) Written results shall be submitted for all tests and shall show the following as a minimum:
 - Run length; and
 - Attenuation.
 - 4) Cables which do not meet performance standards shall be re-terminated or replaced as necessary to meet the standards.

END OF SECTION